OFFLINE SIGNATURE FORGERY DETECTION BASED ON GEOMETRIC MEASURES USING TENSORFLOW MODEL

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***Abstract*—The most popular method for identifying people from past signatures is through signatures. By using a TensorFlow model which is a deep learning algorithm, we created a new system to verify signatures on bank checks and other important documents. The main point is that a signature plays a crucial role in banking because it is necessary for money withdrawals. In banks there are no efficient systems to check if a signature is real or fake as it is the most frequently used bio-metric method to verify a person's identity. This can lead to bank fraud. The Project will make it easier to tell whether a signature is genuine or not. Online and offline verification are the two methods of verification. We're going to use various geometric measurements to accomplish offline verification. Python libraries were employed in this case. In the testing phase the model's performance is evaluated. To protect the integrity and legitimacy of handwritten signatures, signature forgery detection is an essential responsibility. The capacity to recognize forged signatures with accuracy has grown more important as a result of the rise in digital transactions and the growing reliance on electronic documents. The act of duplicating or copying another person's signature with the purpose to deceive or gain illegal access is known as signature forging. Manually spotting these forgeries can be difficult because expert forgers are adept at closely imitating the visual style and traits of real signatures.Therefore, there has been a lot of interest in the development of automated methods and algorithms to identify fake signatures.**

***Index Terms*—IQ score,distance measures,machine learning, and data mining.**

I. INTRODUCTION

Legal papers and financial transactions frequently employ signatures as an authorization technique. Given the aforementioned reality, signatures are very important in commercial settings and the creation of legal documents. Due to the value of a signature, there has been a significant increase in signature fraud in the modern era to commit crimes without the knowledge of authorized people. Signature verification is the procedure used to determine whether a submitted signature is real, i.e., to check it is not a counterfeit signature. Traditional neural networks frequently fail to recognize the distinctive letter connections and stroke patterns found in real signatures. First, we will use a database of signatures to teach the system. The system has previously authorized this database of signatures. To verify other signatures, this system will analyze all signatures and create a reference signature. Between the claimed signature and the signature that will be scrutinized distance is calculated.The signature is recognized as forgery if the distance between the parameters of real signature and verification signature exceeds a predefined threshold.

Here, we have used deep learning [1] and machine learning algorithms for creating a sound signature forgery detection system. These models can automatically extract relevant features from signature images and make accurate predictions, reducing the dependence on human experts and improving the efficiency of forgery detection. The use of signature forgery detection extends to various domains, including banking, legal documents, government identification, and authentication systems. Detecting and preventing signature forgeries not only helps in minimizing financial losses and fraud but also safeguards the reputation and credibility of organizations and individuals. By leveraging advanced machine learning techniques, these detection systems offer a reliable and efficient means to combat forgery attempts, protect individuals and organizations from financial and legal repercussions, and maintain the integrity of important documents.

The process of signature forgery detection involves analyzing various characteristics of the signature to determine its authenticity. This analysis may include the examination of stroke patterns, line quality, pressure, rhythm, proportions, and overall consistency. Furthermore, aspects such as pen lifts, timing, and sequence of strokes may also be considered. In

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TABLE I
TABLE FOR EXPANSIONS

S. No	Abbreviation	Expansion
1.	FAR	False Acceptance Rate
2.	FRR	False Rejection Rate
3.	SURF	Speed up Robust Features
4.	SIFT	Scale Invariant Feature Transform
5.	OSV	Original Seen & Verified
6.	SCNN	Shallow Convolutional Neural Network
7.	DCNN	Deep Convolutional Neural Network
8.	PCA	Principal Component Analysis
9.	GLCM	Gray Level Co-occurrence Matrix
10.	SKPCA	Sparse Kernel PCA
11.	SVM	Support vector Machine

digital signatures, additional factors like metadata, encryption algorithms, and authentication protocols may be examined.

To aid in signature forgery detection, experts utilize a range of techniques and tools. These can include manual examination by forensic document examiners, who possess specialized knowledge and expertise in analyzing signatures. They carefully inspect the physical or digital documents, study the ink or digital traces, and use various magnification and imaging techniques to detect irregularities consistencies. It is worth noting that signature forgery detection is a complex and specialized field, and the outcomes of such analyses are typically considered expert opinions rather than definitive proof [2]. The results are often presented in court proceedings or used in forensic investigations to provide insight and support decision-making processes.

Signature forgery detection is the process of identifying signatures that have been falsified or tampered with. This is a critical task in a variety of settings, such as financial institutions, government agencies, and businesses. TensorFlow provides a number of features that make it well-suited for this task, including Signature forgery detection using TensorFlow is a complex process, but it can be a very effective way to protect against fraud and identity theft. The following table lists a few words and their expansions shown in TableI.

A. Objectives

1) legitimacy Assessment:: The main goal is to evaluate a signature's legitimacy by separating real signatures from forgeries.

2) Accuracy Improvement:: The use of a TensorFlow model aims to enhance the accuracy of signature forgery detection. By leveraging machine learning algorithms and neural networks, the model basically learns from a huge dataset of real and forged signatures allowing it to recognize patterns and subtle cues that might be challenging for human experts to detect.

II. RELATED WORK

The identity of a person is verified using a signature in most scenarios. The process of signature forgery includes a collection of data pre-processing (RGB to gray), extracting features, and validating signatures against training data.

A person's signature is a quick and easy way to confirm their identification. It can be confirmed using neural networks [3] by matching and stored as an image. Online or offline signature verification is also possible. Legendre polynomials are used for analyzing and verifying behavioral biometrics such as handwriting in banks or legal documents.

Signature verification is used as a biometric approach in most legal, financial, and bank sectors. The verification process includes testing the signature against a real signature to assess the genuinity of the signature. In designing a offline signature model the developer might face more issues and concerns.

Signature verification also involves analyzing pen speed, pressure, and motion while taking samples of the signature for verification which provides people with secure systems and services for verification. In this model, OSV (Open Seen and Verified) technique can also be included to verify a customer's identity, address, or phone number.

Legendre polynomial co-efficient approximation [4] makes signature matching or verification convenient by producing fixed-length representations for features and uses SGD with momentum to produce better Equal Error Rate Reduction. In this model, two styles of signatures can be used to evaluate the performance of the verification system namely Western and Chinese. The results conclude that there is a good correlation between verification errors and consistency factors for optimal combination of features.

Developing a Signature forgery detection system using CNN and Capsule neural network [5] to increase the efficiency and accuracy of the system. The hybrid architecture of CNN-CapsNet claims to reduce the complexity of the network, and feature extraction phase, and captures the spatial properties of signature images.

Combining of GLCM and features using SVM [6] for creating a model for Offline Signature Verification System which uses eight geometric features and twenty-two co-occurrences matrices. A skew kurtosis method is also used to select optimal features in the final verification of signatures (forged and genuine). This model also shows much improvement in FAR and FAR parameters compared to other existing Signature forgery Systems.

Offline Signature recognition and verification based on neural networks also include four components such as data acquisition, pre-processing, extracting features, recognition, and verification. In this system MLP (Multilayer perceptron), Modular Neural network, and SOM (Self Organizing maps) [7] are used where SOM groups data into clusters for unsupervised training and also includes analysis of data and visual analysis. The mapping problem of multi-dimensional data is tackled using MLP neural networks that are trained with feed-forward-back propagation. After four component steps testing of the model was performed through FAR, FRR, and TER.

In reality, not every case involving signature verification will proceed to the forensic examination phase. We need to quickly validate signatures in several common application cases, such as business, industry, etc. For instance, when the case is still in the preliminary inquiry stage, investigators typically need to

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conduct a short signature inspection to assess the likelihood or direct their research. The usage of commercial signature verification is an additional illustration. Some businesses even retain a single signature for authentication when taking client convenience into account.

The goal of signature forgery detection using a TensorFlow model is to speed up and automate the procedure. A well-trained TensorFlow model can swiftly examine a large number of signatures, offering consistent and objective assessments, as opposed to manual analysis of signatures, which can be time-consuming and subjective.

Scalability is made possible by the use of a TensorFlow model, which is capable of processing a large number of signatures quickly. When dealing with circumstances that require large amounts of documentation, such as banking institutions, legal proceedings, or governmental organizations, this is especially helpful. The TensorFlow model can help human specialists make decisions by enhancing their knowledge of signature forgery detection. It can provide unbiased perceptions and fact-based evaluations that help forensic document examiners or legal experts make choices about the veracity of signatures.

As new forging methods are developed or as additional data becomes available, the TensorFlow model can be continuously updated and enhanced. The model may adapt to changing forging techniques and improve its detection abilities over time by utilizing continuous research and improvements in machine learning.

The TensorFlow model needs to be taught to be flexible and able to generalize its understanding to recognize various kinds of signature forgeries. It should be able to recognize a variety of forging methods, such as freehand, traced, and manipulated digital signatures.

In conclusion, the goals of signature forgery detection utilizing a TensorFlow model center on enhancing precision, effectiveness, scalability, and flexibility while supplying decision assistance and real-time detection capabilities. The model attempts to improve the overall efficiency of signature forgery detection procedures by utilizing the power of machine learning.

III. Methodology

TensorFlow's mechanism for detecting signature fraud includes a number of crucial phases. Here is a general description of what happens:

A. Data Gathering and Preprocessing:

Gather a dataset of signatures that includes samples of both real and fake signatures. Obtain digital signatures or scan physical documents to convert the signatures to digital format. Resize, normalize, and, if necessary, use noise reduction techniques to preprocess the signature photos.

B. Data Labelling:

Label the signatures in the dataset as authentic or counterfeit using data labeling and annotation. If available, annotate each signature with extra information, such as the type of forgery (e.g., freehand forgery, traceable forgery, digital manipulation).

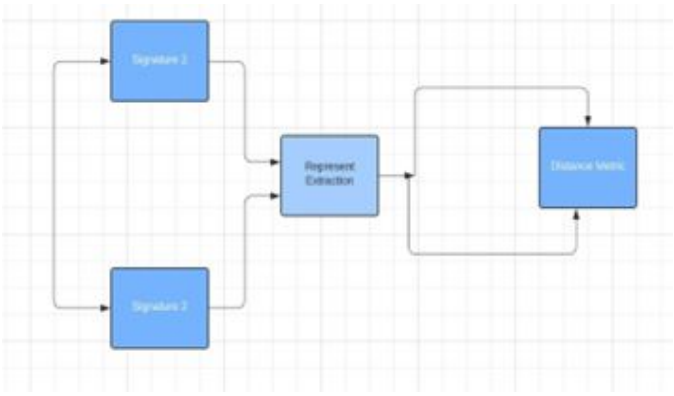


Fig. 1. Workflow model of TensorFlow model

C. Extract

useful features from the signature photos using feature extraction. This might involve a variety of elements, such as proportions, line quality, pressure, rhythm, and general consistency. Pen lifts, time, the order of the strokes, and digital signature metadata are possible extra elements.

D. Architecture design model:

Establish the TensorFlow model's ideal architecture. This may entail utilizing recurrent neural networks (RNNs) [8] for sequential information in signatures and convolutional neural networks (CNNs) for image-based characteristics. As a starting point or set of building blocks for the signature forgery detection model, think of pre-trained models like ResNet, VGG, or Inception. In this model, the following models are used:

1) TensorFlow: TensorFlow is an open source and free software library used mostly in ML (machine learning) models for particularly focusing on deep neural networks and used across many tasks as a math-based library.

2) Matplotlib:: Matplotlib is a plotting library for Python used as a NumPy for numerical mathematics. With the help of a GUI, it provides an object-oriented API for embedding plots into applications.

3) Pandas:: Pandas provide operations and data structures to handle time series and numerical tables. It is mainly used for data manipulation and analysis.

4) Keras:: Keras offers an interface(interface) for ANN (Artificial Neural Networks). It is an open-source library that is also used for data analysis and manipulation [8]. The following figure 1 represents the workflow model of the TensorFlow model.

A person's identity can be shown incontrovertibly and uniquely by their handwritten signature. Due to its simplicity and originality, it holds a crucial position in the field of behavioral bio metrics. Secured authentication is crucial since signatures are the most frequently used bio-metric attribute for verification by law enforcement officers and agencies,

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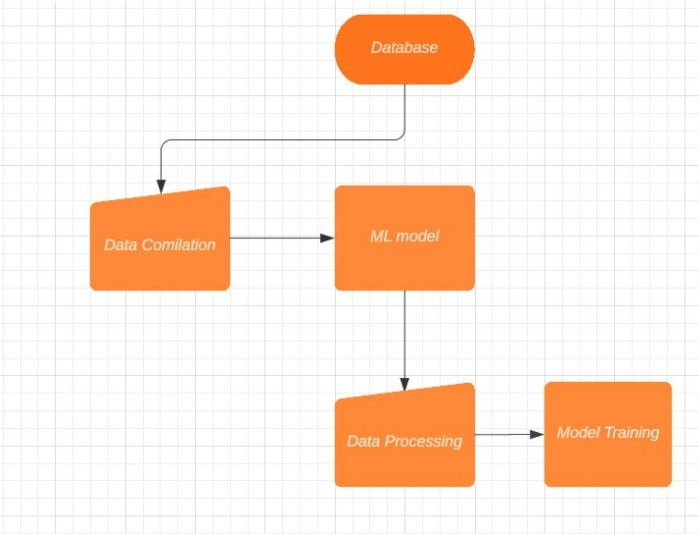


Fig. 2. Flowchart of the TensorFlow Mechanism

particularly in financial institutions, legal transactions, etc. In the era of the digital age, there are a lot of transactions happening where the organizations, like banks, etc., demand handwritten signature verification. In these situations, the signature verification procedure should be quick, accurate, and secure, allowing for real-time verification. In this paper a TensorFlow model along with Pandas, Keras, and Matplotlib libraries. Researchers have used many technologies, such as neural networks and they have done parallel processing to signature verification and they are introducing concepts, new ideas, and algorithms day by day. The following figure 2 represents the Flowchart of the TensorFlow Mechanism.

Gather a sizable dataset of real and fake signatures before creating your dataset. Make sure the dataset is labeled correctly to distinguish between authentic and fake samples.

The basic process of creating a TensorFlow model includes the following steps:

- Data pre-processing: Before training the TensorFlow model, make sure the signature photos are in a format that is appropriate for pre-processing. This can entail scaling the photographs, standardizing the pixel values, and transforming them into a common format.
- Model Architecture: Architecture includes a number of layers for feature extraction, activation functions like sigmoid, tanh, relu based on the requirements and also an FC layer (Fully connected) for the output.
- Training: The training data is inputted into the model and then use an appropriate loss function (such as binary cross-entropy) and optimization technique (like Adam) to refine its parameters. Adjust the weights of the model iterative across a number of epochs to reduce the loss on the training set.
- Evaluation: After feature extraction, model architecture, and lots of training the performance of the model is assessed by validating the training dataset against the new testing data. The evaluation process involves the calculation of precision, accuracy and recall of a model.
- The training dataset's quality: The training dataset should be sufficiently large and representative of the signature kinds that will be found in the actual world. The training dataset should contain only high-quality, accurately classified signatures.
- The model's complexity: The likelihood that a model will attain high accuracy increases with model complexity. However, developing and using more complicated models may also require more computational resources.
- The features chosen: The features that are taken out of the signatures can greatly affect how accurate the model. The features should be able to distinguish between genuine and forged signatures.

When developing and implementing a signature forgery detection system, it is crucial to take into account the variables that may have an impact on the model's accuracy. The following figure 3 represents the pre-processing steps for the raw signature images taken from the dataset.

Typically, to identify signature fraud, a machine learning model is trained to distinguish between real and fake signatures. Such models can be created and trained using TensorFlow, a well-liked deep learning framework.

The following Table II represents the literature review.

IV. Experimental Results:

A model's accuracy is measured by the proportion of times it predicts the outcome accurately. For instance, if a model predicts the outcome 90% of the time, it is 90% accurate. A model's loss represents how well it performs on the training set of data.

The TPR (True Positive Rate) measures how many positive instances are correctly classified, while the FPR (False Positive Rate) measures how many negative examples are wrongly labeled. The ROC curve of a successful model will be towards the top-left corner of the graph.

A. RESULTS OF TENSOR FLOW MODEL FOR SCORE, FRR, FAR:

The score represents the proportion of times the input was correctly detected by the model. The FRR measures how frequently a valid input was mistakenly rejected by the model. The FAR measures how frequently an invalid input was mistakenly accepted by the model. The following table III represents the FAR and FRR score of the TensorFlow Mechanism used.

B. RESULTS FOR TENSOR FLOW MODEL FOR PREDICTED LABEL VS TRUE LABEL:

Each entry in the table corresponds to a sample of the test data. The input's true label is its actual label, whereas the projected label is the one the model assigned to it. The table demonstrates that five times out of five, the model correctly predicted the label. This indicates that the model's accuracy is

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TABLE II
LITERATURE REVIEW

S.No	Title	Approach	Future scope	Method / Technique	Year
1.	Signature Forgery Detection using CNN [1]	CNN	Model can be used and deployed into portable platforms such as Smart Phones and Raspberry Pi's if a signature is forged or real	CNN	2018
2.	Signature Recognition and Forgery detection (Offline) using deep learning models [3]	Harry's Algorithm, surf Algorithm	85 to 89% Accuracy is obtained while detecting the forged signature	Deep Learning	2020
3.	Efficient Signatures verification system based on ANN [8]	Back Propagation using ANN	This model obtains high Accuracy and less error rate	ANN,FAR,FFR	2020
4.	Online Signature verification system using Dl and Legendre polynomial coefficient. [4]	Legendre polynomial coefficients, deep feed-forward neural network	Stochastic gradient descent with momentum deep learning algorithm	Legendre polynomial coefficients, deep feed-forward neural network classifier.	2019
5.	Offline signature verification by fusion of GLCM and geometric features using SVM. [6]	GLCM, HPFI, PCA(SKCPCA),MLYT, GPDS	Grasshopper is a best feature selection technique than compared to other heuristic approaches.	GLCM, HPFI, PCA (SKC PCA), MLYT, GPDS	2020
6.	Signature Forgery detection using SVMs. [2]	CNN	Improvision of CNN by improving the dataset quantity and quality to train a model.	Feature extraction, model training using CNN	2022
7.	Online Signature verification using deep learning descriptors. [9]	SVM, CNN.	Thinking to enhance accuracy by using pre-created filters.	SVM,CNN	2020
8.	Offline Signature recognition and verification based on Neural Networks. [10]	FAR and FRR are reduced by increasing reference sample features.	Multilayer perceptron, modular neural networks.	Used to solve more complicated problems	2013
9.	Offline Signature recognition with deep neural network on a large scale. [11]	C3, RELU	Producing better accuracy for detecting forged signatures.	C3 algorithm, RELU	2019
10.	Offline signature forgery and detection based on the single known sample and deep learning methodologies. [12]	Python image library, deep learning frameworks like tensor flow.	To further optimize the system performance more data is required and can be verified using the voting method.	Meta-Analysis	2020

TABLE III
THE TABLE REPRESENTS THE SCORE, FAR, AND FRR VALUES

Score	FRR	FAR
0.99	0.001	0.001

100 percent. The following table IV represents the predicted label vs the true label.

The formulas and methods are frequently applied in TensorFlow models for the detection of signature forgery. Depending on the model's requirements and architecture, the specific implementation could change.

- Distance-based techniques: These techniques determine how far apart the real and fake signatures are. The likelihood that the signatures are real increases with decreasing distance. The Manhattan distance, the Hamming distance, and the Euclidean distance are a few examples of common distance measures.
- Feature-based techniques: These techniques take characteristics from both real and fake signatures. The attributes can be derived from the signature's shape, the amount of pressure used, or the pace at which it was written [13]. A machine learning model is then trained with the attributes to discriminate between authentic and fake signatures.
- Graph-based techniques: These techniques display the real and fake signatures as graphs. To find discrepancies, the graphs are then put side by side. The hidden Markov model method and the signature graph method are two popular graph-based techniques. The formula that is used relies on the particular signature forgery detection issue. For instance, distance-based approaches are frequently used when the signatures are of good quality and the characteristics are simple to extract. Feature-based approaches are frequently used when the signatures are of poor quality or the characteristics are challenging to extract. Graph-based approaches are frequently employed when the features are complicated or the signatures are of varying quality.

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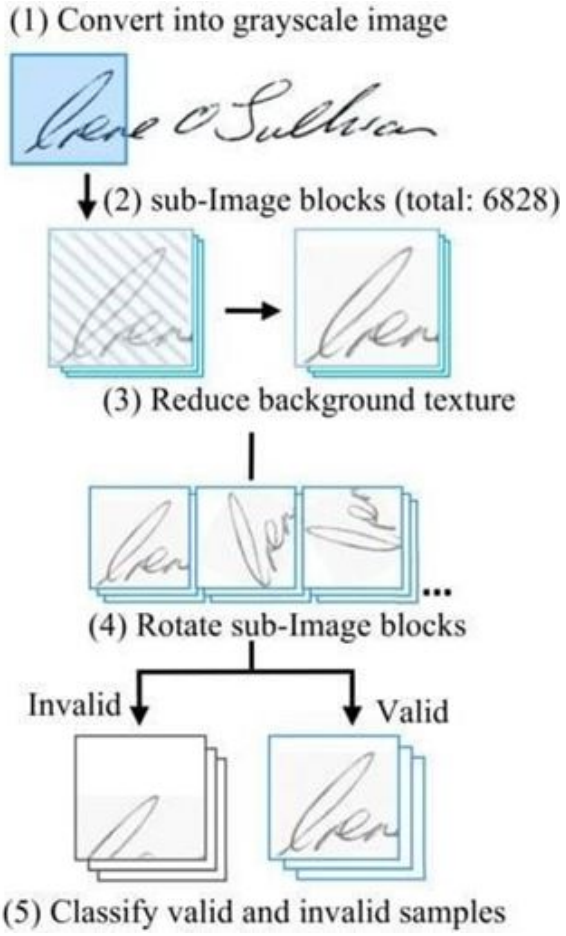


Fig. 3. pre-processing steps for the raw signature images taken from dataset

TABLE IV
PREDICTED LABEL VS TRUE LABEL

S.No.	Predicted Label	True Label
1	0	0
2	1	1
3	0	0
4	1	1
5	0	0

V. CONCLUSION:

In conclusion, detecting signature fraud using a TensorFlow model has enormous promise for stopping fraud and maintaining the reliability of signed papers. For instance, when the signatures are of good quality and the characteristics are simple to extract, distance-based approaches are frequently used. When the signatures are of poor quality or the characteristics are challenging to extract, feature-based approaches are frequently used. When the features are complicated or the signatures are of varying quality, graph-based approaches are frequently employed. Why do we use TensorFlow for this model, it is open source, free to use and is frequently updated with features and functionalities.

The benefits of using a TensorFlow model for signature forgery detection include its capacity to handle complex and non-linear relationships within the data, scalability to process large volumes of signatures effectively, and potential for continuous improvement through retraining with new data. In conclusion, detecting signature fraud using a TensorFlow model is a viable way to stop the spread of fraudulent activity.

VI. FUTURE REFERENCES:

Several of the following areas show potential for future use in signature forgery detection:

- The creation of more complex deep learning designs, such as transformer models and graph neural networks, may help signature forgery detection systems perform more accurately and effectively. These designs are better able to identify fraud because they can capture long-range linkages and hierarchical relationships in signature data.
- Incorporating a variety of data sources, such as pressure data, pen tilt, and biometric traits like finger vein patterns [13] in addition to signature photos, might improve the resiliency of signature fraud detection systems. The fusion of multi-modal data can offer a more thorough and trustworthy analysis, making it more difficult for counterfeiters to imitate real signatures.
- It is increasingly important to establish strong adversarial defense mechanisms as attackers develop advanced methods to get around signature forging detection systems. Adversarial training, in which models are exposed to artificially produced adversarial examples while being trained, can increase the model's resistance to such attacks.
- Constant Improvement and Incremental Updating: Systems for detecting signature forgeries that can incrementally adapt to new data and learn from it will be useful in a dynamic setting. The systems can advance over time and adapt to changing forging strategies by regularly updating the models with fresh signature samples.
- As machine learning models are employed more frequently in crucial fields like signature verification, the capacity to explain and to be interpreted becomes crucial. To train deep neural networks we can also use three types of synthetic data to increase the number of samples and to introduce a new mechanism without the need for additional training. [14]
- Large-Scale Signature Datasets [7]: To create and test signature forgery detection models, it is essential to have access to a variety of comprehensive signature datasets. Large-scale benchmark datasets that include a variety of signatures from many people can promote progress in the field and allow fair comparisons between various methodologies.

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